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Cost-Efficient Study Designs for Binary Response Data with Gaussian Covariate Measurement Error

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SUMMARY

When mismeasurement of the exposure variable is anticipated, epidemiologic cohort studies may be augmented to include a validation study, where a small sample of data relating the imperfect exposure measurement method to the better method is collected. Optimal study designs (i.e., least expensive subject to specified power constraints) are developed that give the overall sample size and proportion of the overall sample size allocated to the validation study. If better exposure measurements can be collected on a sample of subjects, an optimal design can be suggested that conforms to realistic budgetary constraints. The properties of three designs—those that include an internal validation study, those where the validated subsample is derived from subjects external to the primary investigation, and those that use the better method of exposure assessment on all subjects—are compared. The proportion of overall study resources allocated to the validation substudy increases with increasing sample disease frequency, decreasing unit cost of the superior exposure measurement relative to the imperfect one, increasing unit cost of outcome ascertainment, increasing distance between two alternative values of the relative risk between which the study is designed to discriminate, and increasing magnitude of hypothesized values. This proportion also depends in a nonlinear fashion on the severity of measurement error, and when the validation study is internal, measurement error reaches a point after which the optimal design is the smaller, fully validated one.

1. Introduction

Binary response data with continuous covariates, some of which are measured with error, are encountered frequently in epidemiology. For example, standard measures of average daily dietary intake of specific nutrients in nutritional epidemiology are made with considerable error, as documented by the comparison by Willett et al. (1985) of two popular methods of nutritional assessment. Poor physical fitness is an important risk factor for numerous chronic diseases, including cardiovascular diseases, all-cause mortality, and cancer (Blair et al., 1989). When estimated by self-administered physical activity questionnaire, there is evidence that the quantity

Key words: Binary response; Cohort study; Covariate measurement error; Epidemiology; Logistic regression; Study design.

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of interest is observed with substantial error (Kohl et al., 1988). Environmental epidemiologists studying the effects of air pollution, water pollution, and occupational exposures on human health are also plagued by considerable error in the measurement of quantitative exposure variables (Brunekreef, Noy, and Clausing, 1987; Lilienfeld, 1988). Frequently, technology is available to obtain better measurements but the high cost of such methods prohibits widespread use. For example, physical fitness can be directly measured through a maximal effort exercise treadmill test, such as the Balke procedure (Balke and Ware, 1959). Of course, such a procedure cannot often be feasibly implemented on the 10,000 or more study subjects who comprise a large prospective cohort. In nutritional epidemiology, the simple and inexpensive food frequency questionnaire may be regarded as a surrogate for estimates of daily nutrient intake obtained from detailed dietary records. In occupational and environmental epidemiology, exposure measurements obtained from more accessible methods such as work history or distance from a contaminating source serve as surrogates for true personal exposure as might be measured from personal sampling devices worn on the subject's lapel.

There has been a considerable amount of interest among both statisticians and epidemiologists in developing appropriate models for the analysis of this type of data (Carroll et al., 1984; Armstrong, 1985; Stefanski, 1985; Stefanski and Carroll, 1985, 1987; Whittemore and Grosser, 1986; Schafer, 1987; Rosner, Willett, and Spiegelman, 1989). Recognizing the sensitivity of relative risk estimates to assumptions about underlying covariate measurement error, most of the proposed methods for correcting relative risk estimators for bias due to covariate measurement error permit the substitution of estimates of the measurement error model parameters in place of assumptions about these. However, there has not been any investigation of optimal methods for collecting the data that will provide the necessary information for unbiased estimation and valid inference in this setting. The goal of this paper is the development of reasonably inexpensive but statistically powerful cohort study designs for epidemiologic research when a single continuous covariate is measured with error. Along with investigating the general properties of these designs, we wish to demonstrate the feasibility of calculating optimal sample sizes that are tailored to the specific research environment, in the hope that investigators will begin to implement this approach in new work. In Section 2, we describe the logistic–normal measurement error model for which all designs are optimized. In Section 3, we describe three study designs that allow the parameters of a linear measurement error model to be estimated, eliminating the dependency on highly influential but unverified assumptions about these. The proposed designs modify the usual cohort study by including a "validation study," where perfect and imperfect exposure measurements are observed on a small sample of subjects. Sample size formulae that simultaneously consider cost efficiency as well as efficiency with an appropriate statistical inference criterion are provided. Justification for these criteria is given in Section 4. A lack of algorithms to perform the difficult numerical calculations required by models arising in this setting has previously created an additional obstacle for investigators; numerical solutions to this are presented in Section 5. In addition, Section 5 considers important features of the three designs, including percentage allocated to the validation study, overall cost, and relative efficiency comparisons.

2. The Model

Suppose that when the data are collected by the perfect exposure measurement method, the data follow the logistic model

$$f(D \mid X) = \exp\{(\alpha + \beta X)D\} \Big/ \left[1 + \exp\{\alpha + \beta X\}\right], \tag{1}$$

where D is the observed binary outcome (1 = disease present, 0 = disease absent), and X is the covariate measured without error. The parameter α corresponds to the logit of the sample

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disease prevalence at $X = 0$ (the unexposed), and β, the parameter of interest, equals the logit of $\Pr(D = 1 \mid X = x + 1)$ minus the logit of $\Pr(D = 1 \mid X = x)$, which may be used to estimate the risk ratio of the exposure–disease relationship when the disease is sufficiently rare. Other choices for the form of $f(D \mid X)$ are possible (for example, a probit structure leads to computationally simpler results). Because of nearly universal interest in logistic regression as the regression model of choice in epidemiological research, this paper considers only this model. Logistic regression naturally parameterizes the log relative risk, the parameter of interest in research, to give a log-linear dose–response relationship for relative risk, and bears a close relationship to underlying survival models. A generalized Gaussian measurement error model of the form

$$f(X \mid Z) = \exp\left\{-\tfrac{1}{2}(X - \alpha' - \gamma Z)^2/\sigma^2\right\}\Big/\sqrt{2\pi\sigma^2} \tag{2}$$

describes the relationship between the "true" exposure X and the covariate measured with error, Z. The structural model is implicitly adopted throughout this paper, since we assume that X has a well-defined conditional distribution given Z. In observational epidemiology, where it is rarely possible (or ethical) to allocate the sample covariate distribution, this assumption seems reasonable. Note that when X is distributed Gaussian with mean μ_x and variance σ_x^2 that the classical structural measurement error model (Cochran, 1968)

$$Z = X + \epsilon, \quad \epsilon \sim \mathrm{N}(0, \sigma_\epsilon^2) \tag{3}$$

is obtained by setting α' to $(1 - \gamma)\mu_x$, γ to $\sigma_x^2/(\sigma_x^2 + \sigma_\epsilon^2)$, the reliability coefficient, and σ^2 to $(1 - \gamma)\sigma_x^2$.

We make the usual conditional independence assumption imposed upon surrogate variables, $f(D \mid X, Z) = f(D \mid X)$. That is, Z contains no additional information about the exposure–disease relationship after X is taken into account. It is easy to show that this assumption is equivalent to the form

$$f(Z \mid X, D) = f(Z \mid X), \tag{4}$$

which is often referred to in the epidemiologic literature as the nondifferential misclassification or random error assumption. This assumption allows us to construct a model for the observed data

$$f(D \mid Z) = \int_{-\infty}^{+\infty} \left(\exp\{(\alpha + \beta X)D\}\Big/\left[1 + \exp\{\alpha + \beta X\}\right]\right) \cdot \left(\exp\left\{-\tfrac{1}{2}(X - \alpha' - \gamma Z)^2/\sigma^2\right\}\Big/\sqrt{2\pi\sigma^2}\right) dx, \tag{5}$$

which is not analytically evaluable in closed form. However, numerical methods have been developed to evaluate this function, allowing standard likelihood-based methods of inference to be applied as usual (Crouch and Spiegelman, 1990). Alternatively, inference based on approximate estimating equations appears to be a useful option in many instances (Stefanski, 1985; Stefanski and Carroll, 1985, 1987). The pseudo-likelihood approach (Gong and Samaniego, 1981) is another possibility for parameter estimation in this setting.

This paper considers efficient design of cohort studies only—i.e., studies in which the health outcome is random in the design, and exposure is observed at baseline. The logistic–normal model (5) cannot be used for consistent estimation of β in case–control studies under the standard sampling scheme where the numbers of cases and controls in the study are fixed in advance by design, such that the number of controls depends by design on the number of cases. The results of Prentice and Pyke (1979) do not extend to this model, since the logistic structure for $f(D \mid Z)$ is needed for their results to hold. Useful work on parameter estimation in the standard case–control setting has been presented by Armstrong, Whittemore, and Howe (1989).

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Without any data that can be used to model the joint distribution of (X, Z) one may assume explicit values for $(\alpha', \gamma, \sigma^2)$. Since the estimates of (α, β) depend strongly on these assumed values, this is not an attractive approach for scientific research, where much rests on obtaining estimates that can be used to answer a scientific question with confidence. Rather than making unverifiable assumptions about these values, data can often be collected to directly estimate the entire parameter vector. The resulting inference will then account for the full extent of uncertainty surrounding the estimation problem. The remainder of this paper is concerned with designing such studies, with the goal of minimizing financial cost while maximizing efficiency for making the desired inference.

3. Main Study and Validation Study

The designs considered consist of a *main study* of sample size n_1 with observations (D_i, Z_i), $i = 1, \ldots, n_1$, and a *validation study* of sample size n_2 with observations (D_i, Z_i, X_i), $i = 1, \ldots, n_2$, as in an *internal validation study*, or (Z_i, X_i), $i = 1, \ldots, n_2$, as in an *external validation study* (Rosner et al., 1989). The n_1 main study subjects are distinct from the n_2 validation study subjects. The validation study sample size, n_2, will usually be much smaller than n_1, because unit validation study cost will greatly exceed unit cost for the main study. For example, in the Nurses' Health Study, a prospective cohort of over 98,000 women who have been followed since 1976 (Willett et al., 1987), ascertainment of diet through the food frequency questionnaire cost approximately \$1 per study subject, while ascertainment through four 1-week dietary records cost approximately 100 times that amount (Willett, personal communication).

Within this overall framework, the specific design or designs applicable depend on several factors: (1) whether exposure is ascertained *prospectively* or *retrospectively* relative to disease occurrence and disease status assessment, and (2) whether the validation study is *internal*, permitting observation of disease occurrence in this smaller sample, or *external* to the main study, where information concerning only the marginal distribution of $f(X \mid Z)$ can be obtained. When the validation study is external and exposure is prospectively ascertained, design optimization uses the log likelihood function

$$\mathscr{L}_{\mathrm{A}} = \sum_{i=1}^{n_1} \ell_{1i}(\alpha, \beta, \alpha', \gamma, \sigma^2) + \sum_{i=1}^{n_2} \ell_{2i}(\alpha', \gamma, \sigma^2), \tag{6}$$

where $\ell_{1i}(\alpha, \beta, \alpha', \gamma, \sigma^2)$ corresponds to the ith main study subject's likelihood contribution with ℓ_1 the log of (5), and $\ell_{2i}(\alpha', \gamma, \sigma^2)$ corresponds to the ith validation study subject's likelihood contribution with ℓ_2 the log of (2). An example of this design occurs in the study of the effects of physical fitness on chronic disease occurrence, where the validation study compares physical activity data obtained from a self-administered questionnaire to physical fitness measurements obtained from the Balke treadmill test in an independent sample of subjects. In order to use these data, the epidemiologist must be willing to assume that the distribution of $(X \mid Z)$ in the physiologist's data is comparable to that which exists in the main study, an assumption that depends on intrinsic features of the measurement error process and on the underlying marginal distributions of X in the two study populations.

Closely related to this type of design are doubly sampled designs where the validation study is a subset of the main study, and exposure is assessed prior to disease occurrence. Then, design optimization uses the log likelihood function

$$\mathscr{L}_{\mathrm{B}} = \sum_{i=1}^{n_1} \ell_{1i}(\alpha, \beta, \alpha', \gamma, \sigma^2) + \sum_{i=1}^{n_2} \ell_{2i}(\alpha', \gamma, \sigma^2) + \sum_{i=1}^{n_2} \ell_{3i}(\alpha, \beta), \tag{7}$$

where ℓ_3 is the log of (1). Often, ℓ_3 will contribute little information, because the disease is rare and n_2 will be small. For example, in the Nurses' Health Study, the validation study consists of 173 subjects from within the main cohort. Because of the potential for recall bias, it

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may usually be unreasonable to apply $\mathscr{L}_B$ to retrospective cohort studies that rely on self-reports of exposure status or on the subjective judgment of a nonblinded investigator. However, many retrospective cohort studies ascertain exposure status in ways that are far less susceptible to recall bias. For example, in retrospective occupational cohort studies, individual measurements of cumulative exposure to a chemical may be developed by combining work history data obtained from the employer's personnel department with a qualitative scoring system developed by the plant's industrial hygienist. This common method of exposure ascertainment may be validated through personal samplers worn by a subset of cohort members, as in Kreibel et al. (1988).

An additional design that may be useful in this setting is the *fully validated design* (Greenland, 1988a). Rather than collecting data on a large number of subjects with imperfect exposure data, there may be circumstances under which the optimal design consists of a smaller sample of subjects whose exposure status has been "fully validated"—i.e., exposure has been assessed by the superior method only. The use of this design eliminates the measurement error problem entirely, and the usual design and analysis methods are, of course, applicable.

4. Design Optimality Criteria

4.1 *Choosing the Appropriate Inference Criterion*

The goal of most statistical analysis in chronic disease epidemiology is the calculation of a point estimate of relative risk and the construction of confidence intervals around this estimate (Rothman, 1978). A recent paper by Greenland (1988b) compares several criteria that may be used to guide the design of epidemiologic studies geared toward estimation and confidence interval construction. As pointed out by Greenland, determination of study design optimality through the criterion of expected confidence interval width at a specified relative risk has the serious disadvantage of producing designs with unknown, and thus often poor, confidence level for alternative but equally realistic values of the relative risk, and thus is useful only when relative risk is known in advance with a great deal of certainty. The conventional examination of the power of a test for a point null hypothesis also does not guarantee narrow confidence intervals at alternative values.

Instead of these criteria, Greenland proposed using discriminatory power to determine sample sizes. Under the discriminatory power criterion, the optimal design is given by the maximum of two sample sizes calculated by the usual specification of power and size, where the roles of two hypothesized values of relative risk are alternated. That is, the optimal design satisfies the power and size requirements of two hypothesis testing setups: First, β_L plays the role of the null and β_U plays the role of the alternative; then, β_U is the null and β_L the alternative. Here the optimal sample size does not depend on arbitrarily assigning one of (β_U, β_L) to the role of the null hypothesis, and the other to the alternative. Because of the dependence of the variance of $\hat{\beta}$ on the underlying value of β, in models such as (1) and (5), the discriminatory power criterion will lead, in general, to a different optimal design than the one suggested by the traditional power criterion.

Because of its intuitive appeal and lack of dependence on arbitrary factors that more closely parallel the spirit of confidence-interval-based inference, the discriminatory power criterion will be used in what follows below. Note that there is a close mathematical relationship among all of these criteria, and our methods for optimal sample size calculation can be easily modified to apply to these others.

4.2 *Optimality Criteria for Cost Efficiency*

Cost efficiency considerations may be included in the study design in at least three ways. An investigator may wish to identify the optimal allocation of resources (i.e., the optimal values of

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n_1 and n_2) given a fixed budget. In this case, the study's discriminatory power (or other inferential property) is maximized under a fixed cost constraint. In other circumstances, an investigator may wish to identify the optimal sizes of the main and validation studies that provide a desired discriminatory power for the smallest expenditure. This is the criterion on which design considerations will be based in the proposal-writing stage. Finally, an investigator may decide that a validation study is needed after the main study data have been collected. In this case, the additional costs may be minimized by identifying the smallest validation study sample size with which a desired discriminatory power can be attained. Because of its importance in the planning stage, in what follows we will produce optimal designs by minimizing cost subject to a prespecified discriminatory power. Sample size formulae for the other cost efficiency criteria can be easily derived from the methods we have developed.

5. Minimizing Cost for a Specified Discriminatory Power

5.1 *An External Validation Study*

Study design in this setting requires the specification of several nuisance parameters: $\Pr(D)$, the expected sample disease frequency, γ, α', σ^2, parameters from the logistic–normal model of $f(D \mid Z)$, and $\mathrm{E}(Z)$ and σ_z^2, the parameters of the marginal distribution of the imperfectly measured covariate Z. In addition, the investigator must specify the two values of the relative risk, $\exp\{\beta_{\mathrm{U}}\}$ and $\exp\{\beta_{\mathrm{L}}\}$, where $\beta_{\mathrm{U}} > \beta_{\mathrm{L}}$, between which the study will be designed to distinguish with specified discriminatory power π, and the confidence level. Given $\Pr(D)$ and β_{L}, α_{L} is then obtained through the numerical solution to the implicit equation $\mathrm{E}_Z\mathrm{E}_{D\mid Z}(D; \alpha_{\mathrm{L}}, \beta_{\mathrm{L}}, \alpha', \gamma, \sigma^2, \mathrm{E}(Z), \sigma_Z^2) = \Pr(D)$, and similarly for α_{U}. Finally, the unit costs of the main study, $r_D + r_Z$, and the validation study, $r_X + r_Z$, where r_D, r_X, and r_Z are the unit ascertainment costs for the outcome, the superior exposure assessment method, and the imperfect exposure method, respectively, must be specified. Optimal sample sizes, n_1 and n_2, are the maximum of the two values obtained as the solution to

$$\min_{n_1, n_2:\, n_1, n_2 > 0} \left\{ n_1 + n_2 \frac{r_X + r_Z}{r_D + r_Z} \right\}$$

subject to the constraints

$$1 - \Phi\left(\frac{z_{1-\alpha/2}\sqrt{V_{\mathrm{L}}(n_1, n_2; \alpha_{\mathrm{L}}, \beta_{\mathrm{L}}, \alpha', \gamma, \sigma^2)} - \beta_{\mathrm{U}} + \beta_{\mathrm{L}}}{\sqrt{V_{\mathrm{U}}(n_1, n_2; \alpha_{\mathrm{U}}, \beta_{\mathrm{U}}, \alpha', \gamma, \sigma^2)}} \right) \geqslant \pi, \tag{8}$$

$$\Phi\left(\frac{-z_{1-\alpha/2}\sqrt{V_{\mathrm{U}}(n_1, n_2; \alpha_{\mathrm{U}}, \beta_{\mathrm{U}}, \alpha', \gamma, \sigma^2)} - \beta_{\mathrm{L}} + \beta_{\mathrm{U}}}{\sqrt{V_{\mathrm{L}}(n_1, n_2; \alpha_{\mathrm{L}}, \beta_{\mathrm{L}}, \alpha', \gamma, \sigma^2)}} \right) \geqslant \pi, \tag{9}$$

where $z_{1-\alpha/2}$ is the $1 - \alpha/2$ percentile of the standard normal distribution, $1 - \alpha/2$ corresponds to the specified confidence level, and $V_{\mathrm{L}}(n_1, n_2; \alpha_{\mathrm{L}}, \beta_{\mathrm{L}}, \alpha', \gamma, \sigma^2)$ and $V_{\mathrm{U}}(n_1, n_2; \alpha_{\mathrm{U}}, \beta_{\mathrm{U}}, \alpha', \gamma, \sigma^2)$ are the variances of the estimator of β under the two alternative values of log relative risk being considered.

We assume that sample sizes are sufficiently large that an estimator is available that is nearly unbiased, normal, and efficient. Further investigation is needed to determine the minimum sample sizes for which this assumption is realistic for current proposed methods. Because it is now possible to obtain the maximum likelihood estimates of β and the other parameters of model (5) (Crouch and Spiegelman, 1990), the asymptotic inverse Fisher information will be used to approximate the needed variances $V_{\mathrm{L}}(n_1, n_2; \alpha_{\mathrm{L}}, \beta_{\mathrm{L}}, \alpha', \gamma, \sigma^2)$ and $V_{\mathrm{U}}(n_1, n_2; \alpha_{\mathrm{U}}, \beta_{\mathrm{U}}, \alpha', \gamma, \sigma^2)$. Design optimality is based on the variance of $\hat{\beta}$ alone, reflecting common practice in epidemiology where interest is clearly focused on the value of the relative risk, while the other

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parameters represent the effects of features of the data for which there is usually no intrinsic interest. This approach is closely related to the D_s-optimality criterion (Begg and Iglewicz, 1980), which has proven useful in the design of randomized clinical trials for similar reasons.

In the case when the validation study is derived from data *external* to the main study, the information $\mathscr{I}_{n_1+n_2}(\alpha, \beta, \alpha', \gamma, \sigma^2)$ follows from (6),

$$\mathscr{I}_{n_1+n_2}(\alpha, \beta, \alpha', \gamma, \sigma^2) = \left[\left(\sum_{i=1}^{n_1} \mathrm{E}_{D|Z}\left\{\frac{\partial^2 \ell_1(D_i, Z_i; \alpha, \beta, \alpha', \gamma, \sigma^2)}{\partial\theta_r\, \partial\theta_s}\right\} + \sum_{i=1}^{n_2} \mathrm{E}_{X|Z}\left\{\frac{\partial^2 \ell_2(X_i, Z_i; \alpha', \gamma, \sigma_2)}{\partial\theta_r\, \partial\theta_s}\right\}\right)_{rs}\right],$$

where $\theta_r, \theta_s = \alpha, \beta, \alpha', \gamma, \sigma^2$. In this design, when $\theta_r, \theta_s = \alpha$ or β, the second term of the element vanishes. When the appropriate element of $\mathscr{I}^{-1}$ is evaluated at $\beta = \beta_{\mathrm{L}}$, $V_{\mathrm{L}}(n_1, n_2; \alpha_{\mathrm{L}}, \beta_{\mathrm{L}}, \alpha', \gamma, \sigma^2)$ is obtained. We assume that the sample will be randomly drawn from the population under study, so the design should be based on the unconditional distribution of $\hat{\beta}$. Since the data are unconditionally independent and identically distributed, standard arguments give the expression used to evaluate the asymptotic inverse information,

$$(n_1 + n_2)\mathscr{I}^{-1}_{n_1+n_2}(\alpha, \beta, \alpha', \gamma, \sigma^2) \xrightarrow{p} \left[\left(p\,\mathrm{E}_Z\left(\mathrm{E}_{D|Z}\left\{\frac{\partial^2 \ell_1(Z_i; \alpha, \beta, \alpha', \gamma, \sigma^2)}{\partial\theta_r\, \partial\theta_s}\right\} + (1-p)\mathrm{E}_{X|Z}\left\{\frac{\partial^2 \ell_2(Z_i; \alpha', \gamma, \sigma^2)}{\partial\theta_r\, \partial\theta_s}\right\}\right)_{rs}\right)\right]^{-1},$$

where p is the limit, as $n_1 + n_2 \to \infty$, of $n_1/(n_1 + n_2)$.

Numerical methods are needed to evaluate those components of the expected information matrix that cannot be obtained in closed form. These components are linear combinations of six double integrals. The inner integrals are calculated to a theoretical accuracy of 10^{-15} through the algorithm and software developed by Crouch and Spiegelman (1990). Because the integrand functions, which are themselves integrals, required by the outer integration can be severely asymmetric, sharply peaked, and have several points of inflection, numerical quadrature using Simpson's rule for some arbitrarily large limits of integration did not permit numerical solutions to the design specifications considered in the accompanying figures and tables, except in a small number of special cases. Results given by Crouch and Spiegelman suggest that standard Gaussian quadrature techniques would not be a reliable alternative for these calculations. Using the method proposed by Crouch and Spiegelman, an accurate algorithm for the outer integration was developed that rigorously proves the error bounds for the suggested numerical integration. Space does not permit the inclusion of the details of this development, kindly supplied by Dr Crouch; interested readers may obtain this by writing to the first author of this manuscript. Outer integrals are calculated to a theoretical accuracy of 10^{-9}. There are no closed-form solutions to these design optimization problems, and n_1 and n_2 are optimized using IMSL's nonlinearly constrained minimization routine DNCONF (IMSL, 1987).

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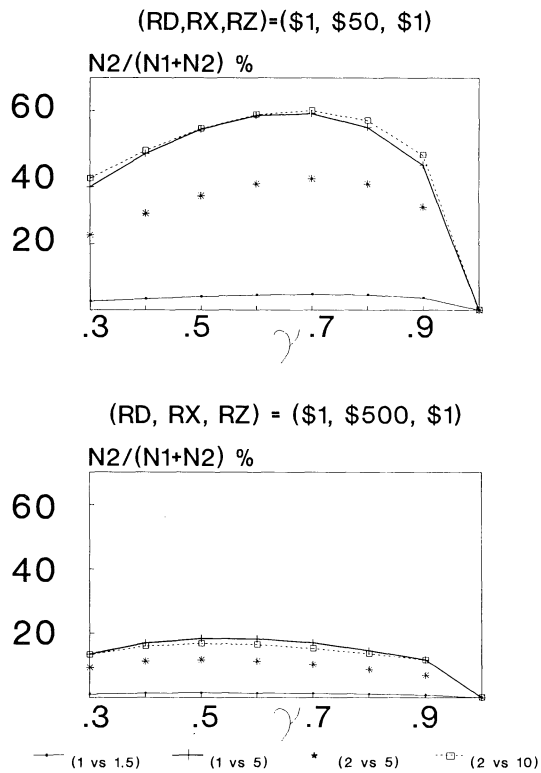

Figure 1. Percent relative external validation study size as a function of the reliability coefficient parameter.

In Figure 1 and all subsequent figures unless otherwise noted, we consider sample disease probabilities corresponding to the extremes of the range seen in common practice in cohort studies, .005–.05, in the case of the classical structural measurement error (3), with $X \sim$ N(0, 1). Note that in the case of no measurement error, $\gamma = 1$. Smaller sample disease probabilities will require sample sizes beyond what is feasible to attain reasonable power, and larger sample disease probabilities indicate violation of the rare disease assumption, where odds ratio inference may no longer be informative if the real interest is in inference about the relative risk. We look at two representative main and validation study unit combinations of unit costs, corresponding to $r_X = \$500$, \$50; $r_Z = \$1$, and $r_D = \$1$, and at several pairs of hypothesized relative risk between which it is a goal of the investigation to discriminate. In this design, it is not necessary to specify the individual unit costs of X, Z, and D, but only the sums $r_D + r_Z$ and $r_X + r_Z$, corresponding to the unit costs of the main study and external validation study. Type I error rate is set at .05, and discriminatory power is .95. We do not force sample sizes to be integer-valued, thereby adding further smoothness to the figures.

Figure 1 shows the relationship between the percent relative validation study size $100[n_2/(n_1 + n_2)]$ as a function of the reliability coefficient parameter γ in the classical structural measurement error model (3), with $X \sim$ N(0, 1), for Pr(D) = .005. We observe that, unless both hypothesized relative risks are near the null value, the relative validation study size rises sharply from 0 as soon as measurement error is introduced into the data, and begins to fall slowly after a point that varies with the design specifications. When the pair of hypothesized relative risks differ little from 1, the optimal validation study size remains small relative to the optimal main study size regardless of the extent of measurement error, reflecting the feature that

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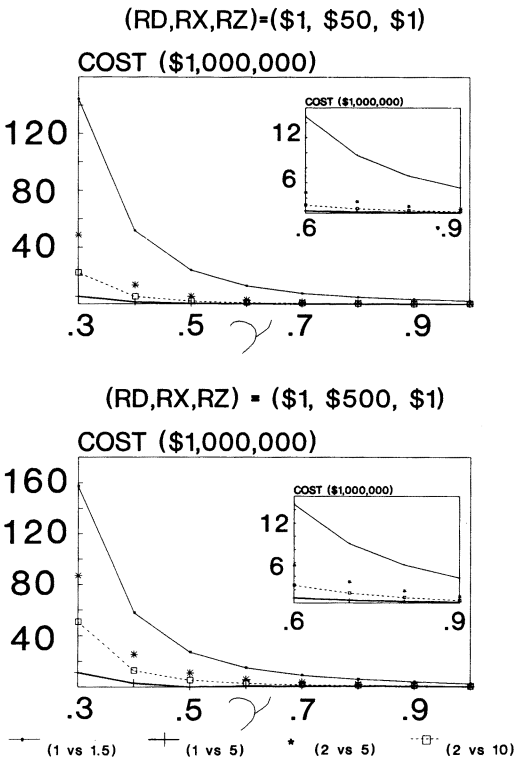

Figure 2. Overall M/EV study cost in dollars as a function of measurement error, for $\Pr(D) = .05$.

near the null, measurement error has relatively little effect. Relative validation study size increases with increasing log distance between the hypothesized relative risks, and appears to depend little on their absolute magnitudes. These patterns remain unchanged for $\Pr(D) = .05$, and beyond (not shown). Proportionately more of the study is devoted to validation as the unit cost ratio of the validation study to the main study decreases, and as the sample disease probability increases (not shown).

Figures 2 and 3 show the relationship between overall main study/external validation study (M/EV) cost in dollars, directly reflecting absolute sample size, and measurement error, for $\Pr(D) = .05$ (Fig. 2) and $\Pr(D) = .005$ (Fig. 3). We see that cost rises dramatically both as measurement error increases and as the log distance between the hypothesized relative risk decreases, except when the disease is extremely rare and the distance between the hypothesized relative risks is large. To a lesser extent, cost increases with the absolute magnitudes of the hypothesized relative risks as well, particularly for larger sample disease probabilities and unit cost ratios. Overall study cost is considerably lessened as the sample disease probability increases. With measurement error of γ less than .6, the cost required to conduct a powerful cohort study investigating a rare disease becomes extremely high, particularly when the hypothesized relative risks are not far apart.

5.2 *An Internal Validation Study*

The validation study in this design contains information about the exposure–disease relationship specified by (1)—i.e., the parameters α and β—as well as the measurement error model. This feature of the internal validation study has a strong influence on the properties of the design, and is responsible for its differences with the external validation study design. Because the parameter

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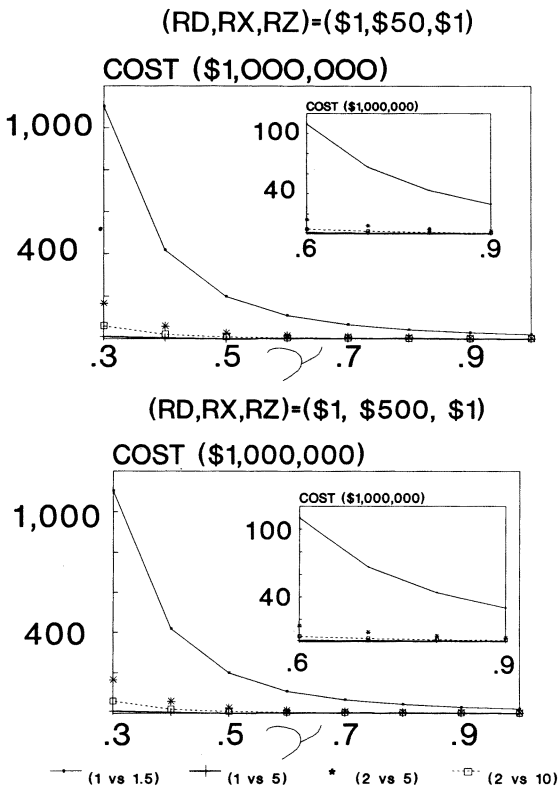

Figure 3. Overall M/EV study cost in dollars as a function of measurement error, for $\Pr(D) = .005$.

of interest, β, can be estimated from the internal validation study alone given sufficient data, optimal design in this setting must consider two competing frameworks: a main study/internal validation study design, and a fully validated design. We find the optimal main study/internal validation study design by again using the expectation of the inverse Fisher information over the sampling distribution of X and Z to derive the variances for β_{U} and β_{L} and minimizing study cost as a function of n_1 and n_2 subject to constraints (8) and (9). After specifying $\Pr(D)$, β_{L} and β_{U}, α', γ, σ^2, μ_x and σ_x^2, α_{L} and α_{U} can then be found as the implicit solutions to the equation $\Pr(D) = \mathrm{E}_X \mathrm{E}_{D|X}(D;\ \alpha,\ \beta,\ \alpha',\ \gamma,\ \sigma^2)$ where $\beta = \beta_{\mathrm{L}}$, β_{U} ard $X \sim \mathrm{N}(\mu_x,\ \sigma_x^2)$. To derive the variances $V_{\mathrm{L}}(n_1,\ n_2;\ \alpha_{\mathrm{L}},\ \beta_{\mathrm{L}},\ \alpha',\ \gamma,\ \sigma^2)$ and $V_{\mathrm{U}}(n_1,\ n_2;\ \alpha_{\mathrm{U}},\ \beta_{\mathrm{U}},\ \alpha',\ \gamma,\ \sigma^2)$ in this case, the information $\mathscr{I}_{n_1+n_2}(\alpha,\ \beta,\ \alpha',\ \gamma,\ \sigma^2)$ follows from (7):

$$
\begin{aligned}
&\mathscr{I}_{n_1+n_2}(\alpha, \beta, \alpha', \gamma, \sigma^2) \\
&= \left[\left(\sum_{i=1}^{n_1} \mathrm{E}_{D|Z}\left\{\frac{\partial^2 \ell_1(D_i, Z_i; \alpha, \beta, \alpha', \gamma, \sigma^2)}{\partial\theta_r\, \partial\theta_s}\right\}\right.\right. \\
&\quad + \sum_{i=1}^{n_2} \mathrm{E}_{X|Z}\left\{\frac{\partial^2 \ell_2(X_i, Z_i; \alpha', \gamma, \sigma^2)}{\partial\theta_r\, \partial\theta_s}\right\} \\
&\quad \left.\left. + \sum_{i=1}^{n_2} \mathrm{E}_{D|X}\left\{\frac{\partial^2 \ell_3(D_i, X_i; \alpha, \beta)}{\partial\theta_r\, \partial\theta_s}\right\}\right)_{rs}\right], \quad \text{where } \theta_r, \theta_s = \alpha, \beta, \alpha', \gamma, \sigma^2. \quad (10)
\end{aligned}
$$

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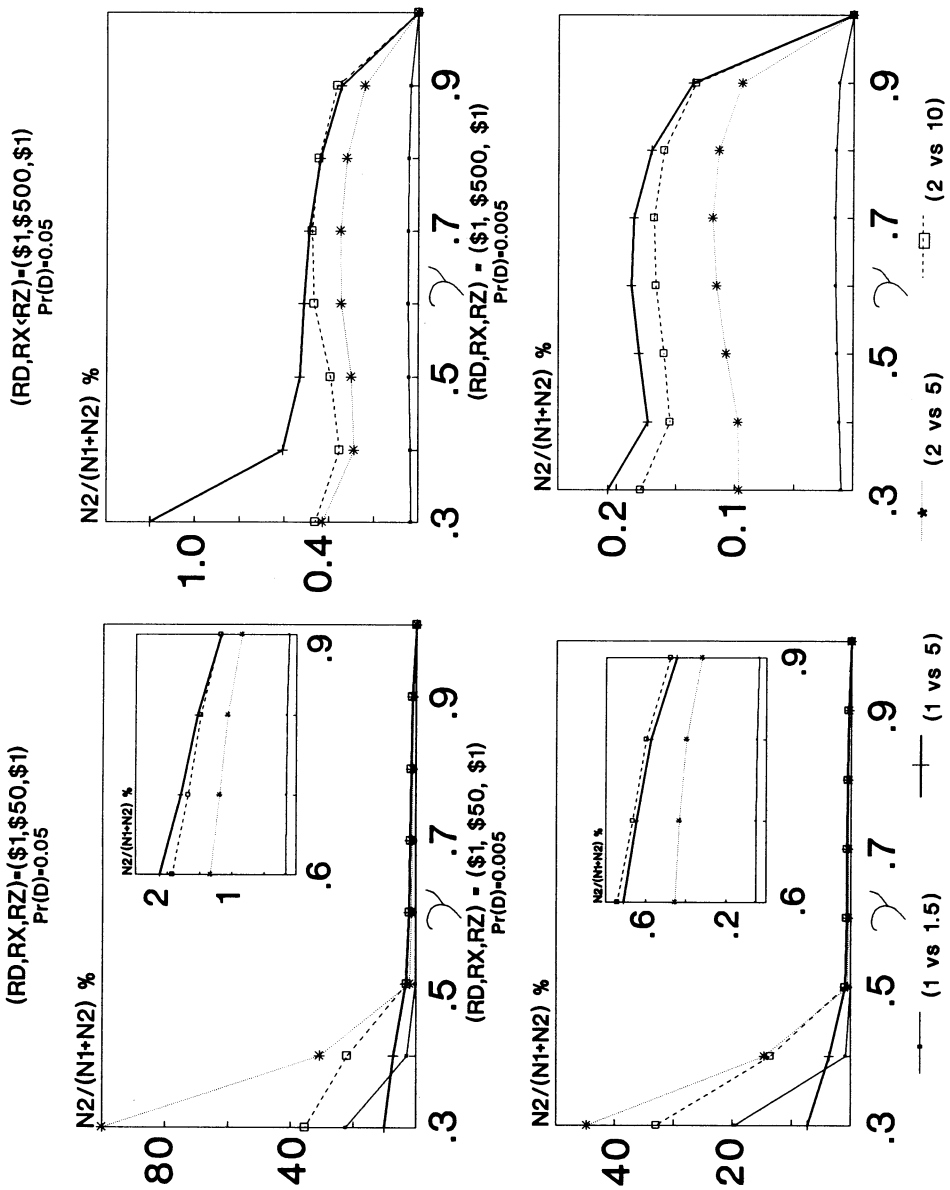

Figure 4. Percent relative internal validation study size as a function of the reliability coefficient parameter.

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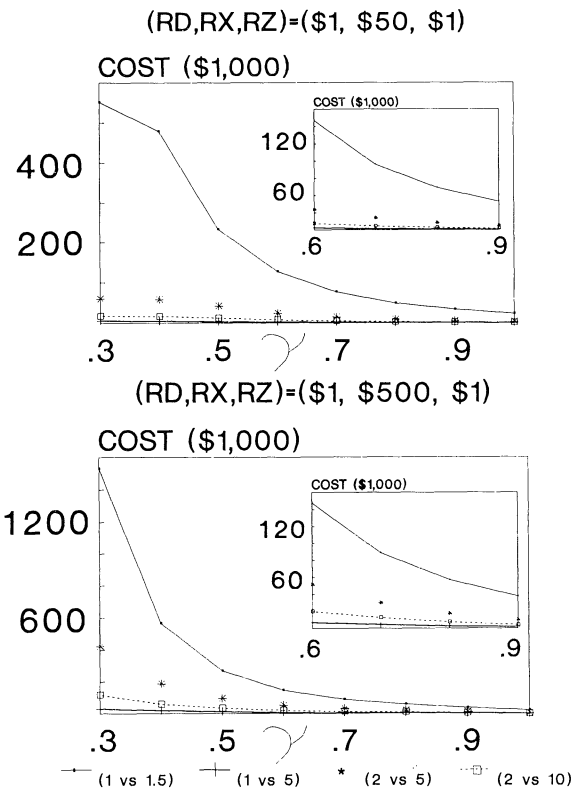

Figure 5. Overall M/IV study cost in dollars as a function of measurement error, for Pr (D) = .05.

Standard asymptotic arguments can again be followed to obtain the convergent values of elements of $\mathscr{I}^{-1}$. The optimal fully validated design is found by identifying the smallest n_2—n_2^*, say—that satisfies constraints (8) and (9), where $V_L(n_1, n_2; \alpha_L, \beta_L, \alpha', \gamma, \sigma^2)$ and $V_U(n_1, n_2; \alpha_U, \beta_U, \alpha', \gamma, \sigma^2)$ are now replaced by $V_L(n_2^*; \alpha_L, \beta_L)$ and $V_U(n_2^*; \alpha_U, \beta_U)$, which are derived from the inverse information of the "true" logistic regression of $D \mid X$ estimating α and β, over the sampling distribution of X. The overall optimal design is then the one with smallest optimal cost, where cost is $(r_D + r_X)n_2^*$ for the fully validated design, and $(r_D + r_Z)n_1 + (r_D + r_Z + r_X)n_2$ for the main study/internal validation study design.

Figure 4 shows the relationship between the percent optimal relative validation study size as a function of the reliability coefficient γ in the classical structural measurement error model (3), with $X \sim$ N(0, 1), for the case where r_D and r_Z are each \$1, and r_X is \$50, \$500. As before, cost is minimized under the constraint that discriminatory power is no less than .95, and the asymptotic confidence level is .95. We see that, unless the ratio of the cost of X to Z is high and the disease extremely rare, when the reliability coefficient gets sufficiently small, the portion of study resources allocated to the validation study begins to increase, and the optimal design is converging to the fully validated one. This appeared to be particularly true when two relatively large hypothesized relative risks were considered. In fact, in the case where Pr(D) = .05, $\beta_L = 2$, $\beta_U = 5$, and r_X = \$50, when γ = .3, the optimal design is the fully validated one. Because this phenomenon has been noted by Greenland (1988a) in the misclassification setting, the conditions under which the fully validated study is optimal or nearly so will be explored in more detail in Section 5.3. Until the measurement error reaches a threshold degree of severity, by comparing Figure 4 to Figure 1 we see that the features of the percent optimal relative main study/internal validation study (M/IV) are similar to features of the optimal validation study

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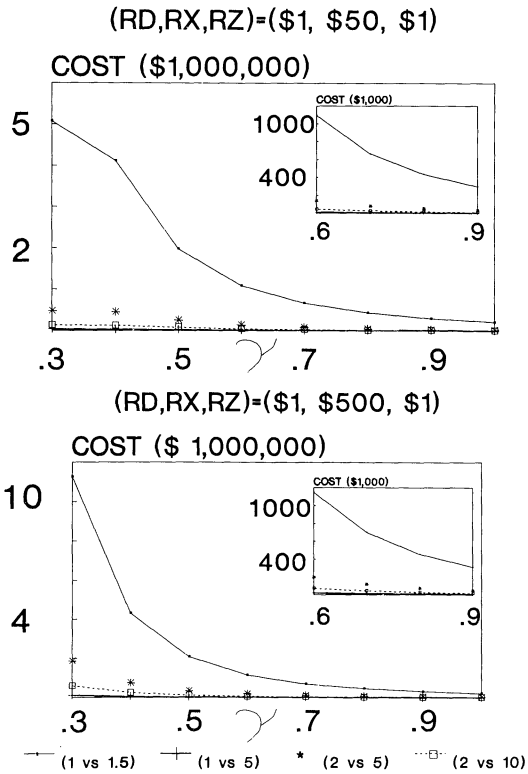

Figure 6. Overall M/IV study cost in dollars as a function of measurement error, for Pr $(D) = .005$.

proportion obtained for the external validation study design: Relative validation study size increases sharply with increasing measurement error, decreasing sample disease probability, decreasing log distance between the hypothesized relative risks, and increases less markedly with increasing absolute magnitudes of the hypothesized relative risks. After a threshold is reached, however, rather than decreasing slowly as seen before, the optimal percent validation study size increases rapidly. In the moderate measurement error region, relative validation study size is considerably smaller in the internal design, due to the additional cost of measuring D.

In marked contrast to analogous optimal external validation study designs, when the sample disease probability is sufficiently large, optimal absolute sample sizes, however, are considerably smaller when the validation study is internal to the main study, as can be seen by comparing Figure 5 to Figure 2, or Figure 6 to Figure 3. This is especially true when measurement error is severe and when the distance between the hypothesized relative risks is small or their absolute magnitudes are large. Except when the distance between the two hypothesized relative risks is large, the internal validation study design is more cost-efficient than the external validation study design. This efficiency advantage is barely perceptible when the sample disease is rare and the cost of D is small relative to the cost of Z (not shown). To summarize, in most cases considered here, the internal validation study design is considerably more efficient as well as more valid. However, in cases where exposure status must be retrospectively assessed, it may not be possible to conduct an internal validation study.

5.3 *Validation Substudy or Fully Validated Study?*

Greenland (1988a) uses an example of a case–control study exploring the association between sudden infant death syndrome and antibiotic use to underscore the point that, in settings similar

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to this example, where a binary exposure variable is subject to nondifferential misclassification, the fully validated design is optimal or nearly so for cost ratios r_X/r_Z in the range of 5 to 10. The case–control ratio in his example is 1:1. Because the multinomial frequency function specified by a discrete misclassification matrix allows a great deal more flexibility in modelling the measurement error process than the homoscedastic, perfectly symmetric, moderate tail weight continuous counterparts considered here, we do not expect results from the misclassification setting to have immediate generalizability to the continuous covariate case. However, past experience suggests that the analogies are often there.

Figure 7 shows the relative cost reduction given by the optimal design, compared with the fully validated one, for $r_X = \$50$, $\$500$, and $r_D = r_Z = \$1$; $r_D = r_X/2$; $r_D = r_X$, and $\Pr(D) = .05$. Relative cost reduction (CE) of the optimal design (OD) is defined as

$$\mathrm{CE} = \left[\$\mathrm{COST}(\mathrm{OD}) - \$\mathrm{COST}(\text{fully validated design})\right] \Big/ \$\mathrm{COST}(\text{fully validated design}),$$

where \$COST(main study/internal validation study design) $= (r_D + r_Z)n_1 + (r_D + r_Z + r_X)n_2$ and \$COST(fully validated design) simply equals $(r_D + r_X)n_2^*$. When $\gamma = 1.0$, $n_2 = 0$ for the main study/internal validation study design. When r_D is relatively low compared to r_X, the relative cost reduction given by the optimal design is greater with larger ratios of r_X to r_Z, and for moderate or large measurement error, it is greater for smaller distances between the hypothesized relative risks. For the studies with larger sample disease probabilities and low unit cost of ascertaining D relative to Z, the optimal design is considerably more efficient up to

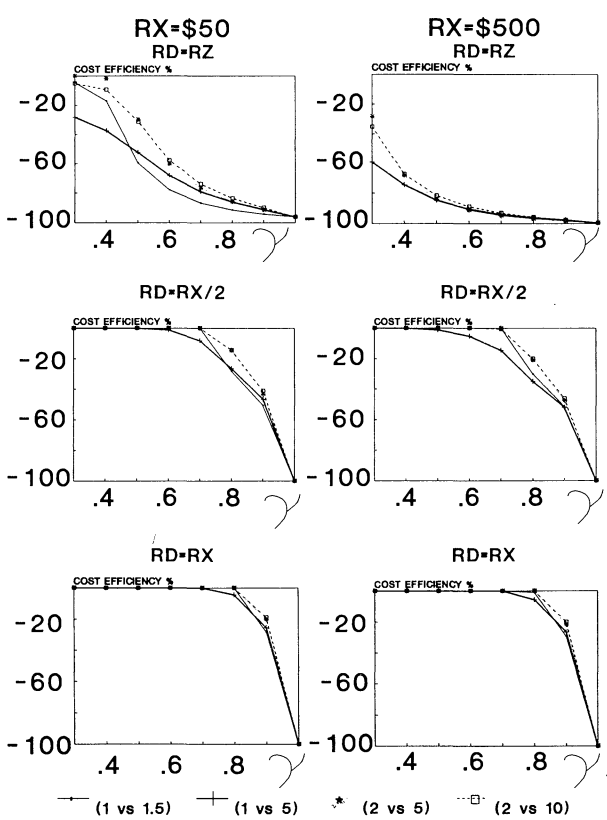

Figure 7. Relative cost reduction given by optimal design, compared with fully validated design, as a function of the reliability coefficient.

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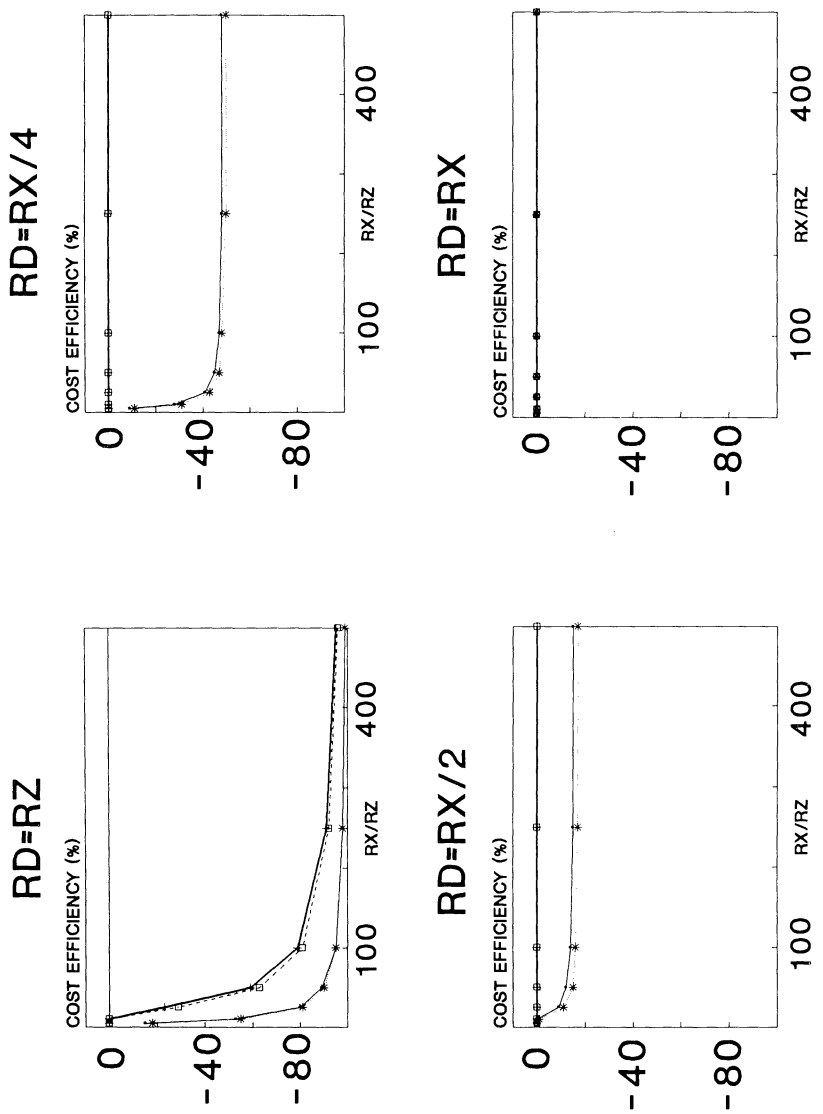

Figure 8. Relative cost reduction given by optimal design, compared with fully validated design, as a function of r_X/r_Z.

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reliability coefficients of .6 or less. Then the cost efficiency of the fully validated design begins to approach that of the optimal, particularly for larger hypothesized relative risks (not shown). For studies of a rare disease ($\Pr(D) = .005$) and a large unit cost ratio, the optimal design is superior to the fully validated design for all degrees of measurement error considered, when r_D is relatively small (not shown). As the ratio r_X/r_Z decreases, the relative difference in cost between the two designs decreases with increasing measurement error. The rate of decrease is enhanced by increasing cost of r_D. In fact, for $r_D = r_X/2$ or larger, the fully validated design is nearly optimal when $\gamma < .7$, over the entire range of sample disease probabilities considered (the case of $\Pr(D) = .005$ not shown), and for all hypothesized relative risks considered. This point should have important implications for study design.

Figure 8 further investigates the dependency of design optimality on r_D, in the important case where $\beta_L = 1.0$ and $\beta_U = 1.5$. We see that, here, design optimality depends little on sample disease probability. For more severe measurement error ($\gamma = .5$), the fully validated design is optimal when $r_D > r_X/4$, and for moderate measurement error ($\gamma = .75$), the optimal design is considerably better than the fully validated design when $r_D < r_X/2$, as long as $r_X/r_Z > 25$. These results are consistent with the suggestion of Greenland, who showed that with exposure misclassification, the fully validated design is an attractive alternative in cases where r_X/r_Z is low and $\Pr(D)$ high. However, it is also clear that there are many important sets of conditions in the cohort study setting where the fully validated design is not a cost-efficient alternative.

6. Discussion

As mentioned in the introduction of this paper, along with exploring the general properties of optimal designs in this setting, we also wish to highlight the availability of the methodology for use in the planning of future investigations in which covariate measurement error is anticipated. In so doing, the investigator must be able to specify a number of quantities: overall study disease frequency, a minimum and maximum hypothesized relative risk between which it is the goal of the study to discriminate, the confidence level, the desired discriminatory power π, the three parameters of the generalized Gaussian measurement error model, α', γ, and σ^2, and the marginal mean and variance of Z, the imperfectly measured exposure variable. Accurate specification of these latter five quantities may be difficult in practice, particularly when the classical measurement error model (3) cannot be assumed, as will frequently be the case. If it is possible to at least specify a range of plausible values for each of these parameters, then this subspace can be informally explored to obtain the "worst-case" optimal design. If so little prior experience has been accumulated that it is not even possible to specify a range of plausible values, then sequential design and analysis procedures will need to be developed to accommodate this situation.

With practical applications in mind, it is important to emphasize that the results presented in this paper pertain only to the classical measurement error model in a single covariate setting. Optimal designs can be obtained with the current methodology for any value of α', γ, and σ^2 ($\sigma^2 > 0$). For example, in the Nurses' Health Study internal validation study, where Z is calorie-adjusted average daily saturated fat intake measured by the food frequency questionnaire and X is the same quantity measured by the average of four 1-week dietary records, $\hat{\alpha}' = 7.256$, $\hat{\gamma} = .468$, and $\hat{\sigma}^2 = 15.04$, $\hat{E}(Z) = 20.7$, $\hat{\sigma}_Z^2 = 23.6$. It does not appear to be worthwhile to explore the properties of optimal designs over the entire five-dimensional parameter space; rather, we encourage investigators to use the available methodology to investigate optimal designs appropriate for their own research environment.

Further work is clearly needed in several areas. First, whether an external or internal validation study is used, design efficiency can be further improved through a sequential approach where the values of the exposure variable Z that are validated are selected from the available data so as to maximize their unit information content. Under the Gaussian modelling assump-

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tions of (2) and (3), optimality should be approximately achieved by sampling from the extremes of the observed distribution of Z appearing in the available data, at least with respect to minimizing the variance of γ because nearly all of the information about this parameter comes from the Gaussian portion of the likelihood function. We might expect that reducing the variance of $\hat{\gamma}$ might have considerable impact upon the variances of estimates of other model parameters, especially $\hat{\beta}$, since under the rare disease assumption correction of $\hat{\beta}$ for bias due to measurement error involves only $\hat{\gamma}$ (Rosner et al., 1989). However, because an optimal allocation rule would be likely to depend heavily on the assumption of normality of the conditional distribution of X given Z, if this assumption is at all violated, the allocation rule could degenerate to a sampling of the outliers.

Related to this point, there is concern about the robustness of these designs to departures from the normality assumption (2). In fact, the observed distributions of many exposure variables in epidemiology appear to be highly nonnormal (heteroscedastic, asymmetric, heavy- or light-tailed), suggesting that the conditional distribution of $X \mid Z$ may share these features. One can usually find appropriate transformations h and h' of X and Z so that $f(h(X) \mid h'(Z))$ does "look" reasonably normal; one must then estimate relative risk in that scale, and transform back to the original scale, which is presumably the one of scientific interest. Of course, this approach has the disadvantage that the linear logistic relationship between disease and exposure in its original scale must be discarded in favor of an assumption of a linear logistic relationship between disease and exposure in the new, "normalized" scale.

A third area for which there may be a considerable applied need is extensions of optimal design methodology to the multiple-covariate case. In addition to the quantities listed above, optimal design of multiple-covariate models requires advance specification of a good deal more "nuisance" quantities, involving the relative risks of the confounders on the disease of interest, the parameters describing their measurement error models, including the covariances between them, if applicable, and the first two moments of their joint distribution. It is unlikely that any investigator can specify even a subset of these in advance with reasonable accuracy. The solution here again seems to be the development of sequential design and analysis methodology; it is not clear, however, whether epidemiologists in the field will be willing to adopt such complex procedures. Even in the simpler environment of multiple risk factors but no covariate measurement error, epidemiologists have been slow to develop and use appropriate sequential design and analysis procedures. The two-stage sampling framework described by Breslow and Cain (1988) may be a useful starting point for developing optimal designs for measurement error models with multiple covariates.

In conclusion, we strongly suggest that epidemiologic investigators consider including a validation study along with the main study when exposure variable measurement error is anticipated. This is likely to be the case in most occupational, environmental, and nutritional epidemiology research settings. Given that technology exists for which better exposure measurements can be feasibly and ethically collected on a small sample of subjects, we have shown that, at least for cohort studies, it will often be the case that an optimal or near optimal design (i.e., the fully validated design) can be suggested that will conform to common budgetary constraints. If possible, an internal validation study is preferable on both validity and efficiency grounds. In certain special cases (studies of a relatively more common disease with severe measurement error and a low cost ratio r_X/r_Z, or when the cost of disease status ascertainment is large relative to the cost of obtaining the imperfect exposure measurement), a small, fully validated study is preferable to other design options. In general, optimal main and validation study sample sizes increase with decreasing sample disease frequency, increasing severity of measurement error, decreasing distance between hypothesized relative risks, and increasing magnitude of hypothesized relative risks. The relative validation study size increases similarly, and also with increasing unit cost ratios r_X/r_Z and r_D/r_Z.

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RÉSUMÉ

Lorsque l'on prévoit qu'une variable d'exposition sera mal mesurée, on peut, pour les études épidémiologiques, inclure une étude de validation où un petit échantillon de données est prélevé d'une part selon la méthode utilisée (et imparfaite) et d'autre part selon une méthode donnée pour meilleure. Des plans d'expérience optimaux (c'est à dire, moins coûteux relativement aux contraintes de puissance) ont été développés et fournissent la taille totale de l'échantillon et la proportion de l'échantillon devant être allouée à l'étude de validation. Si des variables d'exposition meilleures, peuvent être collectées sur un échantillon de sujets, un plan d'expérience optimal qui prend en compte les contraintes de coût, peut être suggéré. Les propriétés de trois plans d'expérience ont été comparées: le premier inclue une étude de validation interne; le second utilise un sous échantillon de validation issu de sujets extérieurs à une première enquête; le troisième utilise la meilleure méthode sur tous les sujets. La proportion des ressources, affectée à l'étude de validation par rapport à l'étude complète, augmente avec l'augmentation de la fréquence de la maladie dans l'échantillon; diminue avec le surcoût unitaire entre la meilleure méthode et la méthode imparfaite; augmente avec la distance entre les deux valeurs du risque relatif de l'alternative (pouvoir de discrimination et augmentation de l'ordre de grandeur des valeurs d'hypothèse). Cette proportion dépend aussi, mais de façon non linéaire, de l'intensité de l'erreur faite sur la mesure, et lorsque l'étude de validation est interne, de l'erreur au point pour lequel le plan d'expérience est le meilleur.

REFERENCES

Armstrong, B. (1985). Measurement error in the generalized linear model. *Communications in Statistics – Simulation and Computation* **14**, 529–544.

Armstrong, B., Whittemore, A. S., and Howe, G. R. (1989). Analysis of case–control data with covariate measurement error: Application to diet and colon cancer. *Statistics in Medicine* **8**, 1151–1164.

Balke, B. and Ware, R. (1959). An experimental study of physical fitness of Air Force personnel. *U.S. Armed Forces Medical Journal* **10**, 675–688.

Begg, C. B. and Iglewicz, B. (1980). A treatment allocation procedure for sequential clinical trials. *Biometrics* **36**, 81–90.

Blair, S. N., Kohl, H. W., Paffenbarger, R. S., et al. (1989). Physical fitness and all-cause mortality. *Journal of the American Medical Association* **262**, 2395–2401.

Breslow, N. E. and Cain, K. C. (1988). Logistic regression for two-stage case–control data. *Biometrika* **75**, 11–20.

Brunekreef, B., Noy, D., and Clausing, P. (1987). Variability of exposure measurements in environmental epidemiology. *American Journal of Epidemiology* **125**, 892–898.

Carroll, R. J., Spiegelman, C. H., Lan, K. K. G., et al. (1984). On errors-in-variables for binary regression models. *Biometrika* **71**, 19–25.

Cochran, W. G. (1968). Errors of measurement in statistics. *Technometrics* **10**, 637–666.

Crouch, E. A. C. and Spiegelman, D. (1990). Evaluation of integrals of the form $\int_{-\infty}^{+\infty} f(t)e^{-t^2}\,dt$. Application to logistic–normal models. *Journal of the American Statistical Association* **85**, 464–469.

Gong, G. and Samaniego, F. J. (1981). Pseudo maximum likelihood estimation: Theory and applications. *Annals of Statistics* **9**, 861–869.

Greenland, S. (1988a). Statistical uncertainty due to misclassification: Implications for validation substudies. *Journal of Clinical Epidemiology* **41**, 1167–1174.

Greenland, S. (1988b). On sample size and power calculations for studies using confidence intervals. *American Journal of Epidemiology* **128**, 231–236.

IMSL (1987). *User's Manual: Math / Library*, pp. 895–902. Houston, Texas: IMSL.

Kohl, H. W., Blair, S. N., Paffenbarger, R. S., et al. (1988). A mail survey of physical activity habits as related to measured physical fitness. *American Journal of Epidemiology* **127**, 1228–1239.

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Kreibel, D., Sprince, N. L., Eisen, E. A., and Greaves, I. A. (1988). Pulmonary function in beryllium workers: Assessment of exposure. *British Journal of Industrial Medicine* **45**, 75–82.

Lilienfeld, D. E. (1988). Changing research methods in environmental epidemiology. *Statistical Science* **3**, 275–280.

Prentice, R. L. and Pyke, R. (1979). Logistic disease incidence models and case–control studies. *Biometrika* **66**, 403–411.

Rosner, B., Willett, W. C., and Spiegelman, D. (1989). Correction of logistic regression relative risk estimates and confidence intervals for systematic within-person measurement error. *Statistics in Medicine* **8**, 1051–1069.

Rothman, K. J. (1978). A show of confidence. *New England Journal of Medicine* **299**, 1362–1363.

Schafer, D. W. (1987). Covariate measurement error in generalized linear models. *Biometrika* **79**, 385–391.

Stefanski, L. A. (1985). The effect of measurement error on parameter estimation. *Biometrika* **72**, 582–592.

Stefanski, L. A. and Carroll, R. J. (1985). Covariate measurement error in logistic regression. *Annals of Statistics* **13**, 1335–1351.

Stefanski, L. A. and Carroll, R. J. (1987). Conditional scores and optimal scores for generalized linear measurement-error models. *Biometrika* **74**, 703–716.

Whittemore, A. S. and Grosser, S. (1986). Regression methods for data with incomplete covariates. In *Modern Statistical Methods in Chronic Disease Epidemiology*, S. H. Moolgavkar and R. L. Prentice (eds), 19–34. New York: Wiley.

Willett, W. C., Sampson, L. Stampfer, M. H., et al. (1985). Reproducibility and validity of a semi-quantitative food frequency questionnaire. *American Journal of Epidemiology* **122**, 51–65.

Willett, W. C., Stampfer, M. H., Colditz, G. A., et al. (1987). Dietary fat and risk of breast cancer. *New England Journal of Medicine* **316**, 22–28.

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